

Question 1 (2 points)

Any NFA having n states can be converted to a DFA having at most (select the smallest possible value)

- A. 2^n states
- B. 3^n states
- C. $2n$ states
- D. n^2 states

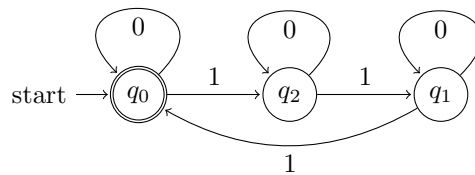
Option A is correct. Refer to the construction of a DFA for a given NFA (Theorem 2 of section 3.2 in Week 1 Lecture Notes). For a given NFA, the states of the constructed DFA correspond to the subsets of states of the NFA. If the NFA has n states, then there are 2^n subsets of states and hence 2^n states in the corresponding constructed DFA.

Question 2 (2 points)

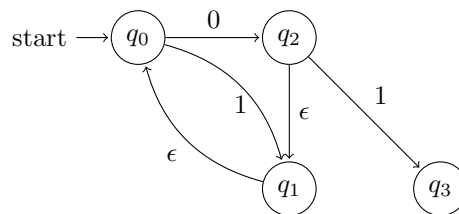
Consider the language $L = \{w \mid \text{the number of 1's in } w \text{ is divisible by 3}\}$ over the binary alphabet $\{0, 1\}$. What is the minimum number of states required for a DFA that accepts L ?

- A. 1
- B. 2
- C. 3
- D. 4

Option C is correct. We require 3 states to count the number of 1's modulo 3. The 3 states in the DFA correspond to the number of 1's in the input string being $(0 \bmod 3)$, $(1 \bmod 3)$ and $(2 \bmod 3)$. The minimal DFA that accepts L is shown below.

**Question 3** (2 points)

What is the ϵ -closure of state q_1 in the following NFA?

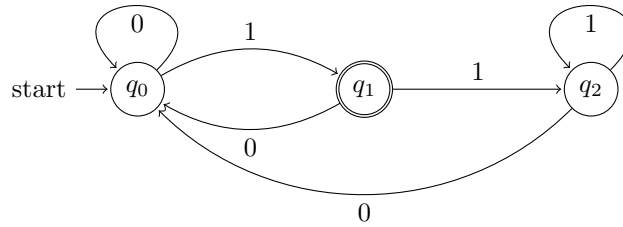


- A. $\{q_0, q_1, q_2\}$
- B. $\{q_0, q_1, q_2, q_3\}$
- C. $\{q_0, q_1\}$
- D. $\{q_1\}$

Option C is correct. ϵ -closure of a state q in an NFA refers to the set of all the states that are reachable from q via only ϵ -transitions.

Question 4 (2 points)

Which of the following strings is accepted by the DFA below?



- A. 01011101
- B. 00111100
- C. 10100011
- D. 11000010

Option A is correct. If we run the DFA on any of the other strings, we reach a state other than the accept state.

Question 5 (2 points)

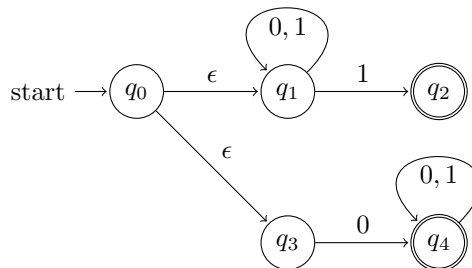
Let L_1 and L_2 be two regular languages. What can we say about the language $L = \{w_1w_2 \mid w_1 \in L_1, w_2 \in L_2\}$?

- A. L may or may not be regular
- B. L is regular
- C. L is not regular
- D. L is accepted by some NFA but there is no DFA that can accept L

Option B is correct. L is the concatenation of the languages L_1 and L_2 . Refer to the Closure Properties of Regular Languages (Section 3.3, Week 1 Lecture Notes)

Question 6 (2 points)

What is the language accepted by the following NFA?



- A. $\{w \in \{0,1\}^* \mid w \text{ ends with 0 and begins with 1}\}$
- B. $\{w \in \{0,1\}^* \mid w \text{ begins with 0 and ends with 1}\}$
- C. $\{w \in \{0,1\}^* \mid w \text{ ends with 0 or begins with 1}\}$
- D. $\{w \in \{0,1\}^* \mid w \text{ begins with 0 or ends with 1}\}$

Option D is correct. It is straightforward to verify that the given NFA has at least one computation path that leads to an accept state if and only if the input string begins with 0 or ends with 1. If the string ends with 1, then there is a computation path for the NFA that leads to state q_2 whereas if the string begins with 0, then there is a computation path for the NFA that leads to state q_4 . On the other hand, any computation path leading to state q_2 indicates that the input string ends with 1 and similarly, any computation path leading to state q_4 indicates that the input string begins with 0.

Question 7 (2 points)

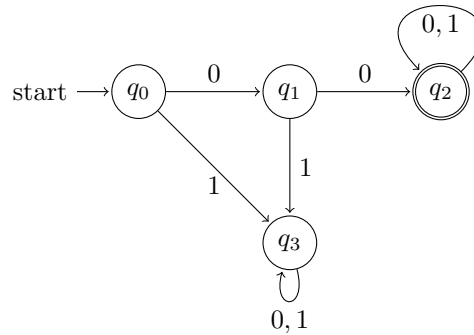
Consider the language

$$L = \{w \in \{0,1\}^* \mid w \text{ begins with } 00\}.$$

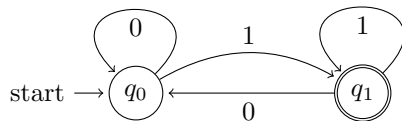
How many states will the minimal DFA for L have?

- A. 2
- B. 3
- C. 4**
- D. 5

Option C is correct. The minimal DFA for L is shown below.

**Question 8** (2 points)

Consider the following DFA.



What is the language accepted by this DFA?

- A. $\{w \in \{0,1\}^* \mid w \text{ has exactly one } 1\}$
- B. $\{w \in \{0,1\}^* \mid w \text{ ends with a } 1\}$**
- C. $\{w \in \{0,1\}^* \mid w \text{ begins with a } 1\}$
- D. $\{w \in \{0,1\}^* \mid w \text{ consists of only } 1\text{'s}\}$

Option B is correct. It is straightforward to verify that the given DFA leads to an accept state if and only if the input string ends with 1. If the string ends with 1, the DFA leads to state q_1 . On the other hand, if the DFA leads to state q_2 , it indicates that the input string ends with 1.